

Week 12 Graded Mini Project

E-Commerce Order, Refund & Exception Handling

Problem Statement

Handle delayed deliveries, refunds, damaged products, return eligibility and exceptional cases through a goal-oriented CrewAI workflow.

Agent Roles

1. Order Issue Identification Agent
2. Policy Interpretation Agent
3. Resolution Recommendation Agent
4. Escalation Agent

Task Flow & Handoffs

Customer Query → Issue Identification → Policy Interpretation → Resolution Recommendation → Escalation Decision

Escalation Logic

Escalate when fraud is suspected, refund exceeds \$500, return window exceeded, confidence below threshold, or urgency is high.

Representative Inputs

Delayed order, damaged product, refund issue, lost shipment, payment fraud, late return request, high-value damaged item.

Expected Outcomes

Routine cases are auto-resolved. High-risk and policy-exception cases are escalated for manual review.

Design Rationale

Each agent performs a focused responsibility and depends on outputs from prior agents, satisfying assignment requirements for collaboration, handoffs and escalation.